Observing the Attractional Effect of Vibrational Frequencies of 2700-2900 Hz on the Eastern Subterranean Termite for Better Capture and Control

Problem

-The Eastern Subterranean termite is a pest that infests 600,000 houses yearly (Jacobs, 2023).

-Termites feed on cellulose: a compound found in paper, cardboard, and wood (Holloway, 2024).

-Termites can cause severe structural damage to infrastructure by consuming wood: estimated to cause around $30 billion in damage in America yearly (Wasson, 2024).

Extermination Methods:

- **Chemical termiticides**: efficient, but have been known to contain harmful chemicals and carcinogens, which could remain in the soil and foundations of the treated property for years (Toxicology, 1982).
- **Bait trap:** poisoned cellulose bait is set up around the termite colony, poisoning the entire colony when consumed. However, this method is very inefficient, since it relies on the termites using the cellulose bait as a food source instead of any other nearby food (Jacobs, 2023).

The bait trap, therefore, could be improved by adding an attractional element to the poisoned bait.

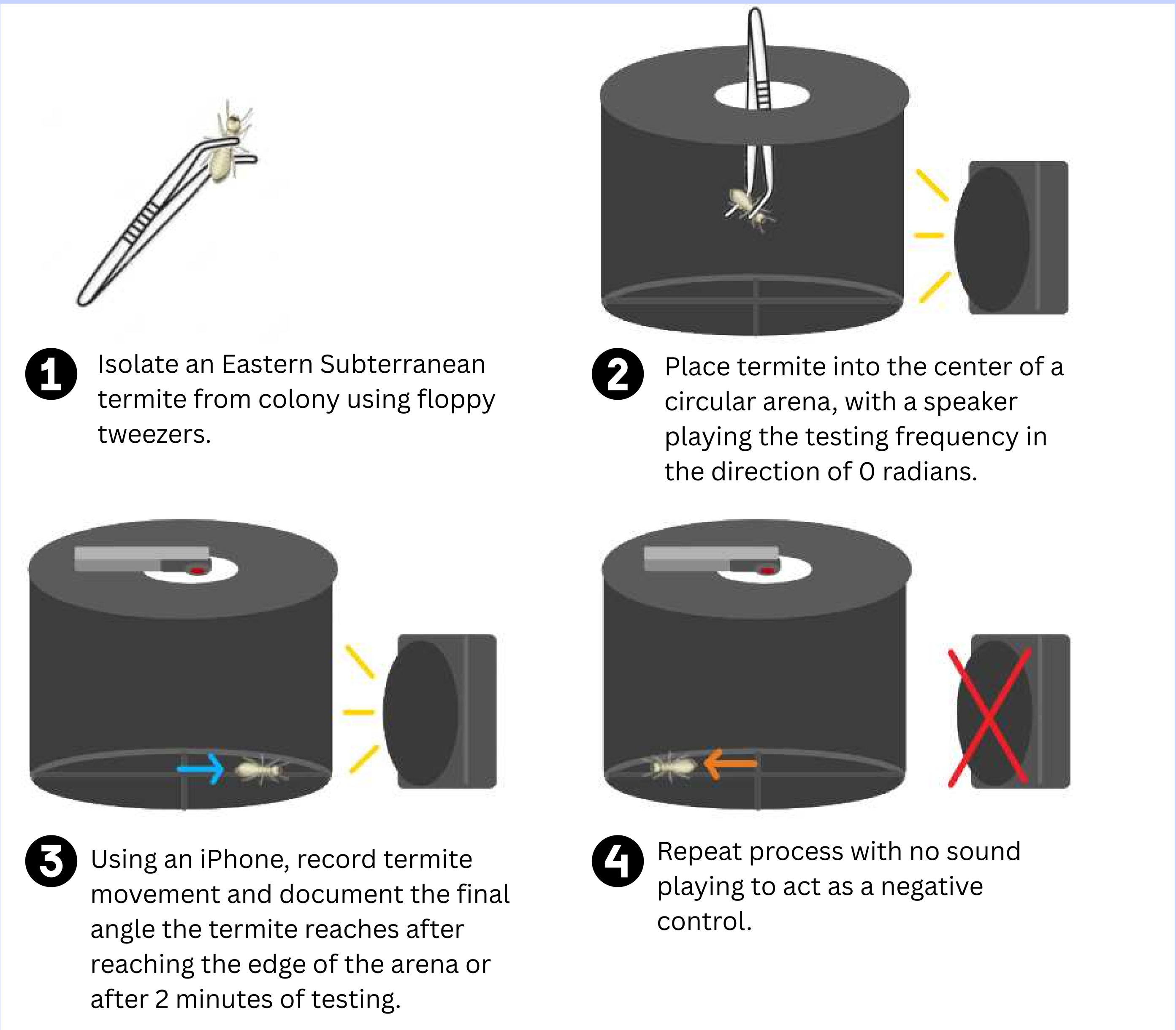
Background & Hypothesis

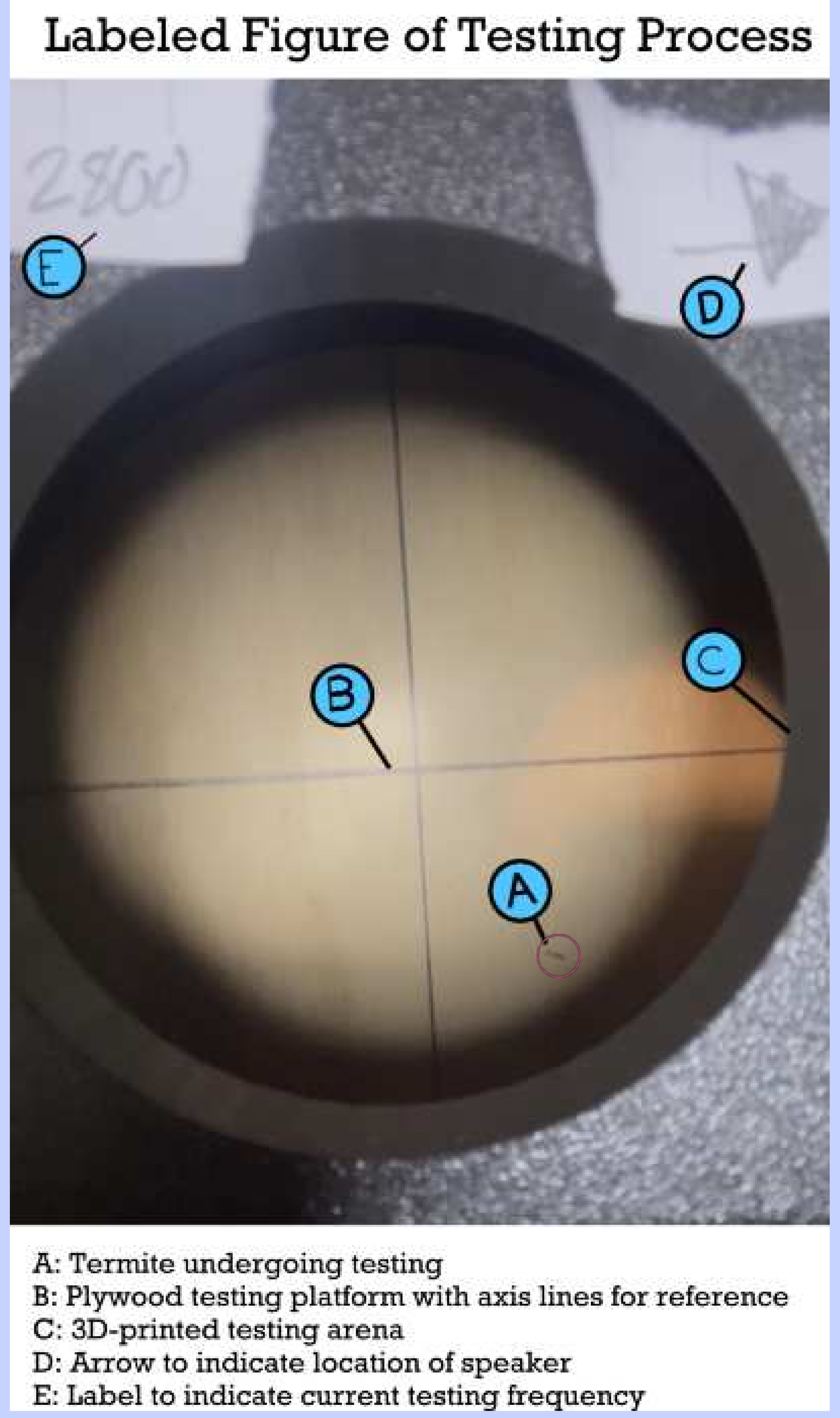
Termites are blind, and are able to find their way around with the help of vibrational signaling, somewhat like echolocation. In a 2005 study, it was found that the Eastern Subterranean termite was attracted to the vibrational frequency of 2800 Hz (Evans et al., 2005).

However, only one recording of a feeding site was analyzed to yield this testing frequency, so a wider range of frequencies should be tested. By finding a “most attractive” frequency, bait traps could be significantly improved.improved.

I believe that frequencies lower than 2800hZ will have the highest attraction rates of the Eastern Subterranean termite, since typically, larger food sources have lower resonant frequencies.

Methodology



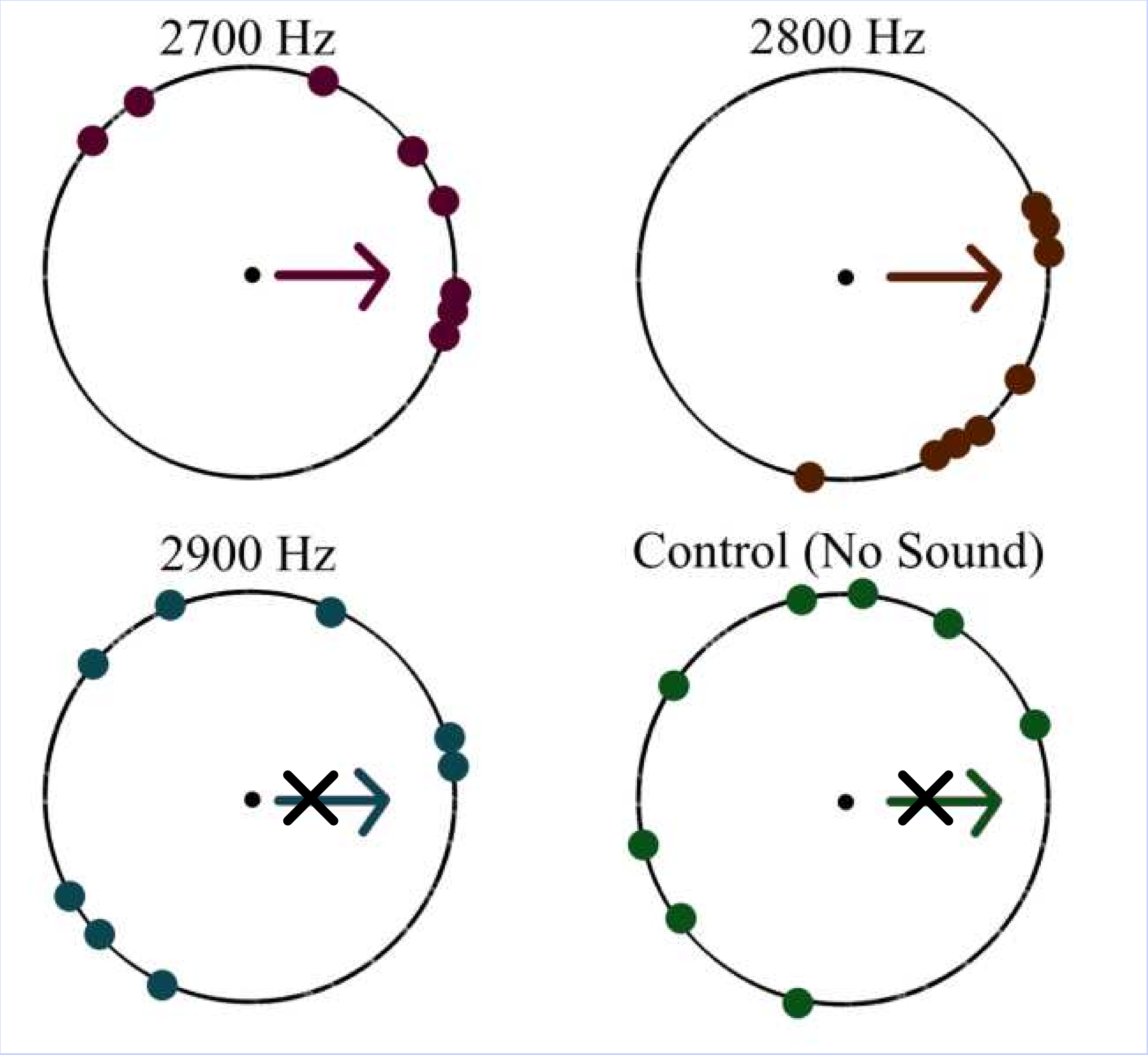


All figures created by me.

IV: the attraction of termites to any tested frequency
DV: the testing frequency played (between 2700-2900 Hz)
Control experiments were run with no sound playing.

32 trials were run in total, with 3 experimental groups (2700, 2800, and 2900 Hz) and one control group (0 Hz).

Results



To interpret collected data, a V-statistical test was run using Phillip Berens’ MATLAB circular statistics toolbox, using the circ_vtest tool (Berens, 2009).

H0= The population is uniformly distributed around the circle
HA=The population is not uniformly distributed around the circle, and has a mean of 0 radians.

The statistical significance level was set at 0.05.

The dots along the circles in the figure represent the final position of tested termites in each testing group.

Conclusion

- Both the 2700 Hz and 2800 Hz experimental groups had a significant mean direction of 0 radians, meaning that these frequencies are significantly attractive to termites (p-values < 0.05).
- The negative control group had values evenly distributed around the circle (p-values > 0.05), meaning that no outside factors significantly affected data distribution.
- The 2900 Hz group had no statistically significant difference from the negative control group, suggesting that it does not attract termites.
- As seen in their V-values, the frequency of 2800 Hz is able to attract termites more consistently than 2700 Hz..

Future Applications

This study suggests that frequencies of 2700 and 2800 Hz are able to significantly attract the Eastern Subterranean Termite, though 2800 Hz frequencies are slightly more attractive. This finding can be used in tandem with bait traps to make them a more viable option for termite control and extermination.

Potentially, future work could involve testing frequencies from 2700-2800 Hz to determine if a frequency in that range results in better attraction, since neither frequency was completely consistent at attracting tested termites.

References

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